

STA256H5 F

Summer 2026 Syllabus

Course Meetings

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Section	Day & Time	Delivery Mode & Location
LEC0102	Tuesday, 1:00 PM - 4:00 PM	In Person: DV 2082
	Thursday, 1:00 PM - 4:00 PM	In Person: DV 2082
	Friday, 11:00 AM - 1:00 PM	In Person: ZZ TBA
LEC0101	Monday, 12:00 PM - 3:00 PM	In Person: DV 2080
	Wednesday, 12:00 PM - 3:00 PM	In Person: IB 150
	Friday, 11:00 AM - 1:00 PM	In Person: ZZ TBA

Refer to ACORN for the most up-to-date information about the location of the course meetings.

Delivery Mode: This course is to be delivered in person as specified on the [University Timetable Builder](#) website. In case there is any change in the mode of delivery, the details will be announced on the course website.

Tutorials and TA Information: All tutorial sessions will be held in person every week from Wednesday May 6th. The details are available on the Quercus page. Tutorials will be administered by the TAs to review topics, solve problems and answer questions. The TAs' contact information and office hours will be posted on Quercus.

Course Contacts

Instructor: Omidali Aghababaei Jazi

Email: omid.jazi@utoronto.ca

Office Hours and Location: Monday & Wednesday, 10:00-11:00 AM. In-person, DH-3086

Instructor: Nishan Mudalige

Email: nishan.mudalige@utoronto.ca

Office Hours and Location: TBA

Course Overview

This course covers probability including its role in statistical modeling. Topics include probability distributions, expectation, discrete and continuous random variables and vectors, distribution functions, distributions of functions of random variables, limit theorems, the central limit theorem.

Course Learning Outcomes

1. Understand and apply fundamental concepts of probability, including sample spaces, events, and some properties of probability functions.
2. Understand discrete and continuous random variables and how to calculate probabilities, expectations, moment generating functions, and learn some inequalities.
3. Understand and apply the concepts and methods for multivariate distributions.
4. Understand some special distributions for both discrete and continuous random variables.
5. Understand concepts of convergence in probability, limiting distribution, and the central limit theorem.

Prerequisites: MAT134H5 or MAT136H5 or MAT137Y5 or MAT139H5 or MAT157Y5 or MAT159H5 or a minimum 65% in MAT133Y5

Corequisites: None

Exclusions: ECO227Y5 or STA257H1 or ECO227Y1 or STAB52H3

Recommended Preparation: MAT232H5 or MAT233H5

Credit Value: 0.5

Course Materials

Textbook: Introduction to Mathematical Statistics (8th Edition) by Rober V. Hogg, Joseph W. McKean and Allen T. Craig, 2019. [Available Here](#)

Additional References

- Probability and Statistics: The Science of Uncertainty (2nd Edition) by Michael J. Evans and Jeffrey S. Rosenthal, 2010.
- Mathematical Statistics with Applications (7th Edition) by Wackerly, Mendenhall and Scheaffer, 2008.

Marking Scheme

Assessment	Percent	Details	Due Date
Term Test	25%	In-person. Friday, May 29th, 2026, 11:00 AM-1:00 PM.	2026-05-29
Quizzes	20%	Weekly (except week 4), Online on Quercus.	Ongoing
Final Assessment	55%	Comprehensive, June (TBA).	Final Exam Period

Quizzes: There will be five quizzes in this course which will be held online. They will generally consist of problems similar to practice problems or examples worked out in class in multiple choice format. Students are expected to complete quizzes by themselves and the only aids allowed for quizzes are their course notes, the textbook, calculators/R. Each quiz will be posted every week, except the 4th week, on Friday at 11:00 AM on Quercus and will be available until the following Sunday at 1:00 PM. The lowest quiz mark will be dropped, and the remaining quizzes will make up 20% of the final mark.

Term Test: The term test will be held in-person on Friday, May 29th, 11:00 AM-1:00 PM. Other details such as the content and the location will be announced on Quercus approximately a week before the test.

Final Exam: The final exam will be in-person, last 3 hours, and from the entire material. The exact date and time will be scheduled and announced by the Office of the Registrar.

Participation Bonus: Students can receive up to a 3% participation bonus by attending lectures (any section) and completing in-class poll questions.

Late Assessment Submissions Policy

Late submissions will **NOT** be accepted.

Course Schedule

Weeks	Topics (textbook sections)	Assessments
Week 1 (May 04 - May 08)	1.1, 1.2, 1.3, 1.4	Quiz 1
Week 2 (May 11 - May 15)	1.5, 1.6, 1.7, 1.8	Quiz 2
Week 3 (May 18 - May 22)	1.9, 1.10, 3.1	Quiz 3
Week 4 (May 25 - May 29)	3.2, 3.3, 3.4, 2.1	Term Test
Week 5 (June 01 - June 05)	2.2, 2.3, 2.4	Quiz 4
Week 6 (June 08 - June 12)	2.5, 2.8, Theorem 2.6.1, 5.1	Quiz 5
Make-up Lecture (June 15)	5.2, 5.3, Review	-

Policies & Statements

Academic Integrity

The Code of Behaviour on Academic Matters states that:

The University and its members have a responsibility to ensure that a climate that might encourage, or conditions that might enable, cheating, misrepresentation, or unfairness is not tolerated. To this end, all must acknowledge that seeking credit or other advantages by fraud or misrepresentation, or seeking to disadvantage others by disruptive behaviour, is unacceptable, as is any dishonesty or unfairness in dealing with the work or record of a student.

It is your responsibility as a student at the University of Toronto to familiarize yourself with, and adhere to, both the Code of Student Conduct and the Code of Behaviour on Academic Matters.

This means, first and foremost, that you should read them carefully.

[Code of Student Conduct](#) and the [Code of Behaviour on Academic Matters](#) are available from the U of T website.

Religious Accommodations

Information about the University's Policy on Scheduling of Classes and Examinations and Other Accommodations for Religious Observances is at <https://www.viceprovoststudents.utoronto.ca/student-resources/rights-responsibilities/accommodation-religious/>

Declaration of Temporary Absence

Students who miss an academic obligation during the term (i.e., in-class assessment, quiz, paper or lab report) may use the ACORN Absence Declaration Tool to record an absence in one or more courses. Students may utilize this option once per term for a single absence period of up to seven consecutive days. The declaration period must include the day of declaration and may include past and/or future dates, for a total of up to 7 calendar days.

Use of the ACORN Absence Declaration does not require supporting documentation and should be used in addition to the missed term work policy outlined in the course syllabus. It remains the student's responsibility to initiate the process for missed academic obligations by following the instructions in the course syllabus.

Re-grading Term Work

A student who believes that their written term work has been unfairly marked may ask the person who marked the work for re-evaluation. Students have up to one month from the date of return of an item of term work to inquire about the mark. If the student is not satisfied with this re-evaluation, they may appeal to the instructor in charge of the course if the work was not marked by the instructor (e.g., was marked by a TA). Such re-marking may involve the entire piece of work and may raise or lower the mark.

For more information on policies regarding re-marking of term work, please refer to [Re-marking Pieces of Term Work in the Academic Calendar](#).

Missed Final Examinations

Students who cannot complete their final examination due to illness or other serious causes must file an [online petition](#) with the Office of the Registrar within 72 hours of the missed examination. Late petitions will NOT be considered. Upon approval of a deferred exam request, a non-refundable fee is required for each examination approved. See the [Office of the Registrar Deferred Examinations](#) for more information.

Accommodations for Students with Disabilities

Students with diverse learning styles and needs are welcome in this course. In particular, if you have a disability/health consideration that may require accommodations, please feel free to approach me and/or Accessibility Services as soon as possible. Accessibility staff (located in room 2240, Student Services Hub, Davis Building) are available by appointment to assess specific needs, provide referrals, and arrange appropriate accommodations. Please call 905-569-4699 or email access.utm@utoronto.ca. The sooner you let us know your needs the quicker we can assist you in achieving your learning goals in this course.

Code of Student Conduct

All students are expected to adhere to the Code of Student Conduct ([Code of Student Conduct \[July 1, 2025\] | The Office of the Governing Council, Secretariat](#)).

Missed Term Work

Missed quizzes will earn a mark of zero, without exception. Reasons/justification will not be accepted.

For missed term test, students must complete and send valid documentation, that is the [Verification of Illness or Injury form](#). The documentation must be sent to the course coordinator (omid.jazi@utoronto.ca) within seven days of the missed term test.

Once per semester, each student is allowed to miss work without any documentation. In that case you must fill out the ACORN absence declaration form. The form can be used at most ONCE per semester (once in total for all your courses, not once per course). The absence you declare can be for a maximum of 7 consecutive days. If you use the ACORN absence declaration form, you do not need to submit any documentation for missed work during that absence.

For a missed term test with valid documentation or absence declaration on ACORN, the weight of the missed term test will be transferred to the final exam.

There will be a make-up test available for students who miss the term test due to a test conflict. The make-up test is optional, students may instead declare their absence on ACORN, in which case the weight of the missed test will be transferred to the final exam.

Academic Integrity

All suspected cases of academic dishonesty will be investigated following procedures outlined in the [Code of Behaviour on Academic Matters \(https://governingcouncil.utoronto.ca/secretariat/policies/code-behaviour-academic-matters-july-1-2019\)](https://governingcouncil.utoronto.ca/secretariat/policies/code-behaviour-academic-matters-july-1-2019). If you have questions or concerns about what constitutes appropriate academic behaviour or appropriate research and citation methods, please reach out to me. Note that you are expected to seek out additional information on academic integrity from me or from other institutional resources. For example, to learn more about how to cite and use source material appropriately and for other writing support, see the U of T writing support website at <http://www.writing.utoronto.ca>. Consult the Code of Behaviour on Academic Matters for a complete outline of the University's policy and expectations. For more information, please see [A&S Student Academic Integrity \(https://www.artsci.utoronto.ca/current/academic-advising-and-support/student-academic-integrity\)](https://www.artsci.utoronto.ca/current/academic-advising-and-support/student-academic-integrity) and the [University of Toronto Website on Academic Integrity \(https://www.academicintegrity.utoronto.ca\)](https://www.academicintegrity.utoronto.ca).

Informed Consent – Email Lists

As a student enrolled in this course, you understand that you are providing your implicit consent to be included in an email list for the department to send you non-essential information from time to time. If you do not wish to be included in such an email list, please request to be removed by contacting one of the Academic Advisors & Undergraduate Program Administrators.

Additional Content

Piazza: Piazza will be used for discussions. This is for student-led discussion. The instructors and the TAs will check Piazza posts on a regular basis and might participate in discussions. Please do not email questions about course content to the instructors or the TAs. Instead, post your question on Piazza. Volume of messages increases one or two days before the test. It will not be possible for the instructors and/or the TAs to answer the questions.

Practice Problems: A list of practice problems will be posted on Quercus. They will form the basis for the term test and the final exam and will be essential to your understanding of the topics covered in class. You are encouraged to work together in groups on practice problems to solidify your knowledge of the material. You are also encouraged to ask your questions during the instructors/TAs' office hours and Piazza.

Email Policy: Email is most appropriate for personal questions. Before you send an e-mail, make sure that you are not asking for information that is already on the course outline/website/announcements, or questions about the course material that are more appropriately discussed during office hours. If you do not get a response, this may be why. If your question is conceptual and does not require calculations or an elaborate answer, you can ask by email. Any questions regarding the tutorials should be addressed to the TAs. For all other matters, contact the instructors. Please email the instructors and the TAs using your U of T email address. The subject line should contain the course number, lecture section number, and a relevant subject (indicating what the email is about). Be sure to include your full name and student number in the body of the message. You will not get a response if you email from other email addresses or do not follow the email policy.

Privacy and Use of Course Materials Notifications: Course materials belong to your instructor, the University, and/or other source depending on the specific facts of each situation and are protected by copyright. In this course, you are permitted to download session materials for your own academic use, but you should not copy, share, or use them for any other purpose without the explicit permission of the instructors. For questions about recording and use of videos in which you appear please contact your instructor.

Use of Generative AI: Students may use artificial intelligence tools, including generative AI, in this course as learning aids. However, students are ultimately accountable for the work they submit.

Academic Commitment: Students are expected to enroll in this course with the understanding that it requires consistent attention and engagement. While we aim to support all learners, the course coordinator cannot modify requirements in response to individual personal circumstances. If you find that the course is not progressing as hoped, please consider reaching out to an academic advisor to discuss possible options, including whether it may be best to withdraw and return to the course at a time that is better suited to your situation.

Students Responsibilities:

- It is up to students to know all course policies and important dates. It is also up to them to know about any important announcements; Check the Quercus page regularly!
- Students are responsible for their own learning. The instructors/TAs are happy to help them learn, but in the end, it is up to them! Use office hours and Piazza often. Make an appointment with the instructors. Keep asking questions until you are satisfied. Ask about big concepts or small details there is no such thing as a stupid question. Always take advantage of extra help and don't wait until it is too late!
- Students must follow the U of T code of Behaviour this means that cheaters will be prosecuted. The Academic Regulations of the University are outlined in the Code of Behaviour on Academic Matters. You are expected to be familiar with, and to abide by, all components of the Code of Behaviour on Academic Matters. Full details can be found online here.

Instructors/TAs Responsibilities:

- Lectures will be clearly presented, organized, and have plenty of examples. Tutorials, office hours, and Piazza can help you solve problems and solidify your learning.
- Extra help, remedial and acceleration, is always available, during office hours, by appointment, and on Piazza.
- Students' emails will be answered in a timely fashion, typically within 48 weekday (business) hours.
- Every student in the class will be treated with fairness and respect. Students who wish to excel are encouraged to consult regularly with the instructors. Students who abuse the U of T code of behavior will be dealt with appropriately.

Course Topics: This course plans to cover the following topics from the textbook.

Chapter 1: Probability and Distributions

1.1 - Introduction

1.2 - Sets

1.3 - The Probability Set Function

1.4 - Conditional Probability and Independence

1.5 - Random Variables

1.6 - Discrete Random Variables

1.7 - Continuous Random Variables

1.8 - Expectation of a Random Variable

1.9 - Some Special Expectations

1.10 - Important Inequalities

Chapter 2: Multivariate Distributions

2.1 - Distribution of Two Random Variables

2.2 - Transformations: Bivariate Random Variables

2.3 - Conditional Distributions and Expectations

2.4 - Independent Random Variables

2.5 - The Correlation Coefficient

2.8 - Linear Combinations of Random Variables

- Theorem 2.6.1: Moment Generating Function of a Linear Combination of Mutually Independent Random Variables

Chapter 3: Some Special Distributions

3.1 - The Binomial and Related Distributions

3.2 - The Poisson Distribution

3.3 - The Gamma, Chi-squared, and Beta Distributions

3.4 - The Normal Distribution

Chapter 5: Consistency and Limiting Distributions

5.1 - Convergence in Probability

5.2 - Convergence in Distribution

5.3 - Central Limit Theorem